



MSc Huali Xu

Doctoral Researcher

Center for Machine Vision and Signal Analysis (CMVS)

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EDUCATION

•University of Oulu

Oulu, Finland

Doctoral Student in Computer Science and Engineering

Sep. 2021 – Present

– Supervisor: Prof. Li Liu

– Research Topic: Computer Vision, Few-shot Learning, Cross-domain Few-shot Learning

•Inner Mongolia University

Hohhot, China

Master's Degree in Computer Science and Technology

Sep. 2018 – June 2021

– Supervisor: Prof. Xiangdong Su

– Master's Theses Title: Research on DNN Image Inpainting Model Guided by Structural Information

•Shanghai Polytechnic University

Shanghai, China

Bachelor's Degree in Computer Science and Technology

Sep. 2012 – June 2016

RESEARCH VISITING

•Nanyang Technology University

Singapore

College of Computing and Data Science

Aug. 2024 – Dec. 2024

– Supervisor: Prof. Hanwang Zhang

– Research Topic: Computer Vision, Cross-domain Few-shot Learning

•Nankai University

Tianjin, China

College of Computer Science

Oct. 2023 – Nov. 2023

– Supervisor: Prof. MingMing Cheng

– Research Topic: Computer Vision, Cross-domain Few-shot Learning

RESEARCH PUBLICATIONS

Huali Xu, Li Liu, Shuaifeng Zhi, Shuzhou Sun, Tianpeng Liu, MingMing Cheng. Step-wise Distribution Alignment Guided Style Prompt Tuning for Source-free Cross-domain Few-shot Learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE TPAMI)*, Under review).

Huali Xu, Shuaifeng Zhi, Shuzhou Sun, Vishal M. Patel, Li Liu. Deep Learning for Cross-Domain Few-Shot Visual Recognition: A Survey. *ACM Computing Surveys* 2025.

Huali Xu, Li Liu, Shuaifeng Zhi, Zhuo Su, Yongxiang Liu, Mingming Cheng. Enhancing Information Maximization with Distance-Aware Contrastive Learning for Source-Free Cross-Domain Few-Shot Learning. *IEEE Transactions on Image Processing (IEEE TIP)* 2024.

Huali Xu, Shuaifeng Zhi and Li Liu. Cross-Domain Few-Shot Classification Via Inter-Source Stylization. *IEEE International Conference on Image Processin (ICIP)* 2023.

Shuzhou Sun, **Huali Xu**, Yan Li, Ping Li, Bin Sheng, Xiao Lin. FastAL: Fast Evaluation Module for Efficient Dynamic Deep Active Learning using Broad Learning System. *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)* 2023.

Huali Xu, Xiangdong Su, Meng Wang, Xiang Hao and Guanglai Gao. An Edge Information and Mask Shrinking Based Image Inpainting Approach. *IEEE International Conference on Multimedia and Expo (ICME)* 2020.

Huali Xu, Xiangdong Su, Tongyang Liu, Pengcheng Guo, Guanglai Gao, Feilong Bao. A Natural Scene Text Extraction Approach Based on Generative Adversarial Learning. *International Conference on Neural Information Processing (ICONIP)* 2019.

Xiangdong Su, **Huali Xu**, Yanke Kang, Xiang Hao, Guanglai Gao and Yue Zhang. Improving Text Image Resolution using a Deep Generative Adversarial Network for Optical Character Recognition. *International Conference on Document Analysis and Recognition (ICDAR)* 2019.

Xiangdong Su, **Huali Xu**, Yue Zhang, Yanke Kang, Guanglai Gao, Batusiren. An End-to-End Preprocessor Based on Adversarial Learning for Mongolian Historical Document OCR. *Pacific Rim International Conference on Artificial Intelligence (PRICAI)* 2019.

Other 3 co-authored papers are shown in my [Google Scholar](#).

RESEARCH PROJECTS

•Super-resolution Reconstruction Based on Medical Images

Supported by Graduate Education Innovation Program of Inner Mongolia Autonomous Region

Hohhot, China

Jan. 2020 – Dec. 2020

- Conduct survey of literature.
- Build experimental framework and experiment.
- Writing of related reports.

TECHNICAL SKILLS

Languages: English, Chinese

Developer Tools: Pytorch, VS Code, Linux,...

Program Languages: Python, C, C++

ACHIEVEMENTS

Academic Activities:

- 2023 IEEE International Conference on Image Processing (ICIP 2023)
- 16th Pacific Rim International Conference on Artificial Intelligence (PRICAI 2019)
- 15th International Conference on Document Analysis and Recognition (ICDAR 2019)

Honor & Awards:

- Excellent Master Dissertation Award of Inner Mongolia Autonomous Region. 2021
- National Scholarship. 2019, 2020

Reviewer:

- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
- Pattern Recognition Letters
- The Vision Computer (TVCJ)
- International Conference on Neural Information Processing (ICONIP)

Teaching Experience:

- Teaching Assistant in Deep Learning Course at University of Oulu. 2021, 2022, 2023